

ELECTRICITY MCQ

1. Electricity is defined as the flow of:

- a) protons through conductors
- b) free electrons from atom to atom
- c) neutrons in nucleus
- d) positive charges only?

Correct answer: b

2. Electric current constitutes motion of charges through:

- a) insulators only
- b) conductors like copper wire
- c) vacuum
- d) gases only?

Correct answer: b

3. SI unit of electric current is:

- a) volt
- b) ohm
- c) ampere when 1 coulomb flows in 1 second
- d) watt?

Correct answer: c

4. Electric circuit requires all except:

- a) battery source
- b) conductor path
- c) switch control

d) vacuum path?

Correct answer: d

5. Conventional current flows from:

a) negative to positive terminal

b) positive to negative terminal

c) randomly

d) no specific direction?

Correct answer: b

6. Electric potential at a point is work done to move unit charge:

a) from infinity to that point

b) within atom

c) from positive to negative

d) against gravity?

Correct answer: a

7. Potential difference V between A and B is:

a) work done per unit charge

b) charge per work

c) current times resistance

d) power divided by time?

Correct answer: a

8. One volt equals:

a) 1 joule per coulomb

b) 1 ampere per ohm

- c) 1 watt per ampere
- d) 1 coulomb per second?

Correct answer: a

9. Ohm's Law states at constant temperature:

- a) I proportional to $1/V$
- b) V proportional to I
- c) R varies with V
- d) P constant?

Correct answer: b

10. Resistance R of conductor equals:

- a) V times I
- b) V divided by I
- c) I divided by V
- d) I times V ?

Correct answer: b

11. SI unit of resistance is:

- a) ampere
- b) volt
- c) ohm when 1A flows at 1V
- d) watt?

Correct answer: c

12. Electrical resistivity ρ measures:

- a) ability to conduct current

b) opposition to current flow

c) power consumption

d) voltage drop?

Correct answer: b

13. Resistance R proportional to:

a) length L and inversely to area A

b) inversely to L and proportional to A

c) L squared

d) A squared?

Correct answer: a

14. Nichrome has highest resistivity making it ideal for:

a) conductors

b) insulators

c) heating elements

d) switches?

Correct answer: c

15. Electrical conductivity σ equals:

a) 1/resistivity

b) resistivity squared

c) resistance times length

d) voltage per current?

Correct answer: a

16. Conductors have:

- a) high resistivity low conductivity
- b) low resistivity high conductivity
- c) zero resistance
- d) infinite resistance?

Correct answer: b

17. Insulators like rubber have:

- a) low resistivity
- b) high resistivity low conductivity
- c) high conductivity
- d) zero resistivity?

Correct answer: b

18. In series connection same quantity flows through all resistors:

- a) voltage
- b) current
- c) power
- d) energy?

Correct answer: b

19. Equivalent resistance R_s in series equals:

- a) sum $R_1 + R_2 + R_3$
- b) $1/(1/R_1 + 1/R_2)$
- c) product $R_1 R_2$

d) average resistance?

Correct answer: a

20. In parallel connection same across all resistors:

a) current

b) voltage

c) resistance

d) power?

Correct answer: b

21. Equivalent R_p in parallel for three resistors:

a) $R_1 + R_2 + R_3$

b) $1/(1/R_1 + 1/R_2 + 1/R_3)$

c) $(R_1 R_2 R_3)^{1/3}$

d) $R_1 R_2 R_3$?

Correct answer: b

22. Heating effect occurs because current through conductor produces:

a) light

b) heat due to resistance

c) magnetism only

d) sound?

Correct answer: b

23. Joule's law $H = I^2 R t$ states heat proportional to:

a) current only

b) square of current resistance time

c) voltage only

d) $1/\text{resistance}$?

Correct answer: b

24. Electric bulb filament uses tungsten because:

a) low melting point

b) high melting point doesn't oxidize easily

c) low resistivity

d) high conductivity?

Correct answer: b

25. Electric fuse melts due to:

a) low resistance

b) excess current heating wire

c) high voltage

d) magnetic field?

Correct answer: b

26. Electric power P equals:

a) VI

b) V/I

c) IR

d) I/R ?

Correct answer: a

27. SI unit of power watt defined as:

- a) 1 joule/second
- b) 1 volt ampere
- c) both a and b
- d) 1 ampere ohm?

Correct answer: c

28. Commercial unit of energy consumption:

- a) watt hour
- b) kilowatt hour kWh
- c) joule
- d) ampere hour?

Correct answer: b

29. 1 kWh equals:

- a) 1000 watt hour
- b) 3.6×10^6 J
- c) both a and b
- d) 1 joule?

Correct answer: c

30. Domestic wiring uses three wires except:

- a) live red
- b) neutral black
- c) earth green

d) blue positive?

Correct answer: d

31. Potential difference between live and neutral:

a) 110V

b) 220V

c) 440V

d) 100V?

Correct answer: b

32. 5A circuit used for:

a) air conditioners

b) fans lights TV

c) geysers

d) irons?

Correct answer: b

33. 15A circuit for high power appliances like:

a) bulbs

b) AC geysers

c) fans

d) mobiles?

Correct answer: b

34. Overloading occurs when:

a) too many devices same circuit exceed capacity

b) voltage too high

c) resistance zero

d) current low?

Correct answer: a

35. Short circuiting low resistance connection causes:

a) low current

b) excessive current flow

c) no current

d) high voltage?

Correct answer: b

36. Earthing provides:

a) high resistance path

b) low resistance path to earth for leakage

c) increases voltage

d) stores charge?

Correct answer: b

37. LED stands for:

a) Light Emitting Diode

b) Light Energy Device

c) Low Energy Diode

d) Light Electric Display?

Correct answer: a

38. Seven segment display uses LEDs to show:

a) letters only

b) decimal numerals

c) colors

d) images?

Correct answer: b

39. LED bulbs advantage over incandescent:

a) short life

b) 90% more energy efficient

c) produce heat UV

d) cannot dim?

Correct answer: b

40. LED TV uses LEDs for:

a) main display

b) backlight in LCD

c) sound

d) power supply?

Correct answer: b

41. Electron flow direction in circuit:

a) positive to negative

b) negative to positive

c) both directions

d) no flow?

Correct answer: b

42. Ammeter measures:

- a) voltage
- b) current
- c) resistance
- d) power?

Correct answer: b

43. Voltmeter connected:

- a) series
- b) parallel across points
- c) through resistor
- d) ground?

Correct answer: b

44. Rheostat used to vary:

- a) voltage
- b) current by changing resistance
- c) power
- d) frequency?

Correct answer: b

45. In series if one resistor fails:

- a) circuit continues
- b) entire circuit breaks
- c) parallel works

d) voltage doubles?

Correct answer: b

46. Parallel advantage:

a) one component failure affects others little

b) high total resistance

c) same current all

d) voltage different?

Correct answer: a

47. For two equal R in parallel R_p equals:

a) $2R$

b) $R/2$

c) R

d) $4R$?

Correct answer: b

48. Series-parallel net resistance calculated first:

a) parallels then series

b) series then parallels

c) randomly

d) multiply?

Correct answer: a

49. Nichrome alloy used because:

a) low resistivity

b) high resistivity high melting point

- c) good conductor
- d) cheap?

Correct answer: b

50. Electric iron uses:

- a) bimetallic strip thermostat
- b) manual switch only
- c) no heating
- d) LED?

Correct answer: a

51. Fuse wire material:

- a) copper high melting
- b) low melting point like zinc
- c) nichrome
- d) tungsten?

Correct answer: b

52. Power $P = I^2R$ used in:

- a) Joule's heating
- b) voltage calculation
- c) current only
- d) resistance?

Correct answer: a

53. 100W bulb at 220V draws current:

- a) 220A

b) 0.45A approx

c) 100A

d) 2.2A?

Correct answer: b

54. Two 10Ω in series equivalent:

a) 20Ω

b) 5Ω

c) 100Ω

d) 10Ω ?

Correct answer: a

55. Three 6Ω parallel equivalent:

a) 18Ω

b) 2Ω

c) 6Ω

d) 3Ω ?

Correct answer: b

56. Domestic meter records in:

a) amperes

b) kWh

c) volts

d) ohms?

Correct answer: b

57. Earth wire prevents:

- a) overloading
- b) electric shock from leakage
- c) short circuit
- d) high voltage?

Correct answer: b

58. Live wire colour in domestic wiring:

- a) green
- b) black
- c) red
- d) yellow?

Correct answer: c

59. Distribution box separates circuits into:

- a) 5A and 15A ratings
- b) single circuit
- c) DC AC
- d) high low voltage?

Correct answer: a

60. LED colours produced using compounds like:

- a) sodium chloride
- b) gallium arsenide phosphide
- c) copper oxide

d) iron sulphide?

Correct answer: b

61. Seven segment displays used in:

a) TV screens

b) digital clocks calculators

c) radios

d) fans?

Correct answer: b

62. LED merit no heat emission means:

a) safe to touch

b) produces UV

c) high energy use

d) short life?

Correct answer: a

63. First LED TV commercial by:

a) Samsung 2005

b) Sony 2009

c) LG 2010

d) Philips 2000?

Correct answer: b

64. Monochrome TV used:

a) RGB LEDs

b) white LEDs

- c) no LEDs
- d) LCD only?

Correct answer: b

65. Resistivity unit:

- a) ohm
- b) ampere
- c) ohm-metre
- d) siemens?

Correct answer: c

66. Conductivity unit siemens per:

- a) ohm
- b) metre
- c) volt
- d) ampere?

Correct answer: b

67. Temperature effect on conductors resistance:

- a) decreases
- b) increases
- c) constant
- d) zero?

Correct answer: b

68. Semiconductors resistivity between:

- a) conductors insulators

- b) only conductors
- c) only insulators
- d) superconductors?

Correct answer: a

69. Superconductors have:

- a) high resistivity
- b) zero resistivity
- c) infinite resistivity
- d) negative?

Correct answer: b

70. Variable resistor symbol represents:

- a) fixed resistor
- b) rheostat
- c) capacitor
- d) inductor?

Correct answer: b

71. Battery symbol shows:

- a) one long one short line
- b) circle
- c) triangle
- d) square?

Correct answer: a

72. Switch symbol:

- a) gap in line
- b) continuous line
- c) zigzag
- d) rectangle?

Correct answer: a

73. In Ohm's law experiment graph V vs I is:

- a) parabola
- b) straight line through origin
- c) hyperbola
- d) circle?

Correct answer: b

74. Slope of V - I graph gives:

- a) current
- b) resistance
- c) voltage
- d) power?

Correct answer: b

75. Copper better conductor than aluminium because:

- a) higher resistivity
- b) lower resistivity
- c) higher density

d) colour?

Correct answer: b

76. Alloys generally have higher resistivity than:

a) pure metals

b) insulators

c) semiconductors

d) liquids?

Correct answer: a

77. Electric energy $W = VIt$ equals:

a) power

b) joules

c) kWh

d) amperes?

Correct answer: b

78. Horsepower equals approximately:

a) 100W

b) 746W

c) 1000W

d) 500W?

Correct answer: b

79. Overload protection by:

a) earthing only

b) fuses MCBs

- c) resistors
- d) capacitors?

Correct answer: b

80. Short circuit prevented by:

- a) high resistance
- b) fuses blowing
- c) increasing voltage
- d) decreasing current?

Correct answer: b

81. LED operates at:

- a) high voltage
- b) low voltage
- c) AC only
- d) DC high current?

Correct answer: b

82. Dimming possible in:

- a) incandescent only
- b) LEDs
- c) both
- d) neither?

Correct answer: c

83. Gallium arsenide LED emits:

- a) red light

- b) blue light
- c) green light
- d) white light?

Correct answer: a

84. Seven segment common types:

- a) common anode cathode
- b) only anode
- c) only cathode
- d) no common?

Correct answer: a

85. Domestic wiring parallel connection advantage:

- a) same current all appliances
- b) each appliance independent voltage
- c) high total resistance
- d) series failure?

Correct answer: b

86. Neutral wire completes:

- a) circuit path
- b) safety only
- c) voltage supply
- d) earthing?

Correct answer: a

87. Electric meter connected:

- a) series with live
- b) parallel
- c) earth only
- d) neutral only?

Correct answer: a

88. TPN distribution board has:

- a) two three neutral
- b) three phase neutral earth
- c) single phase
- d) DC?

Correct answer: b

89. MCB advantage over fuse:

- a) can reset
- b) cheaper
- c) higher melting
- d) no protection?

Correct answer: a

90. Joule's heating maximum when:

- a) R minimum
- b) I maximum R given
- c) V minimum

d) t minimum?

Correct answer: b

91. Bulb glows due to:

a) conduction

b) incandescence heating filament

c) chemical reaction

d) friction?

Correct answer: b

92. Bimetallic strip in iron uses:

a) different expansion rates

b) same expansion

c) magnetic

d) electric?

Correct answer: a

93. Fuse rating matches:

a) voltage

b) normal operating current

c) power

d) resistance?

Correct answer: b

94. CFL advantage over LED:

a) longer life

b) mercury content

- c) higher efficiency
- d) no heat?

Correct answer: c

95. Resistivity alloys like constantan:

- a) varies with temp
- b) nearly constant
- c) zero
- d) infinite?

Correct answer: b

96. Thermistor resistance varies with:

- a) current
- b) temperature sharply
- c) voltage
- d) power?

Correct answer: b

97. LDR light dependent resistor:

- a) resistance increases light
- b) decreases with light
- c) constant
- d) heat dependent?

Correct answer: b

98. Diode allows current:

- a) both directions

b) one direction only

c) AC only

d) blocks all?

Correct answer: b

99. Transistor used for:

a) resistance

b) amplification switching

c) power supply

d) display?

Correct answer: b

100. IC integrated circuit contains:

a) single transistor

b) many transistors single chip

c) resistors only

d) capacitors only?

Correct answer: b